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Installation Survey Circuit Diagrams

Chassis Model	Sales designation	Electrical equipment Version	Installation Chassis end No.	Series/SA	Remarks Group Page
435.111 .115	U 1700 L U 1300 L	Electrical circuit diagram IR wiring, masked circuit and 24 V system	to 078 674	SA 35 769/11	1.5/2
		Circuit diagram IR wiring, masked circuit and 24 V system	from 078 675	SA 35 769/11	1.5/6

Technical Data**Battery**

Battery	SA 35 769/5	SA 35 979
Voltage volt	24	24
Capacity Ah	2 X 55	2 X 125
Low-temperature test current A	255	625
Electrolyte level above upper edge of plate mm	10 – 12	
Charging current A	at initial charge	max. 5%
	normal during recharging	* max. 10% of battery capacity
	during fast charge	to 75%
Maximum temperature	40 ° C	
Maximum temperature	Tropics	50 ° C
Freezing point	well charged	–70 ° C (for tropics– 40 ° C)
	half charged	–25 ° C (for tropics– 13 ° C)
	discharged	–8 ° C (for tropics– 6°C)

* at commencement of gassing falling to 5%

Electrolyte density (normal)

The operating condition of the battery can be approximately determined by the electrolyte density.

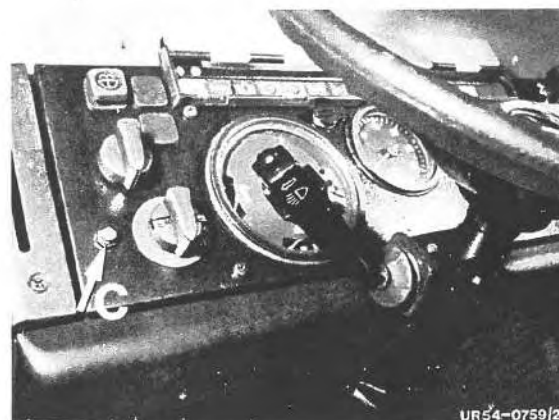
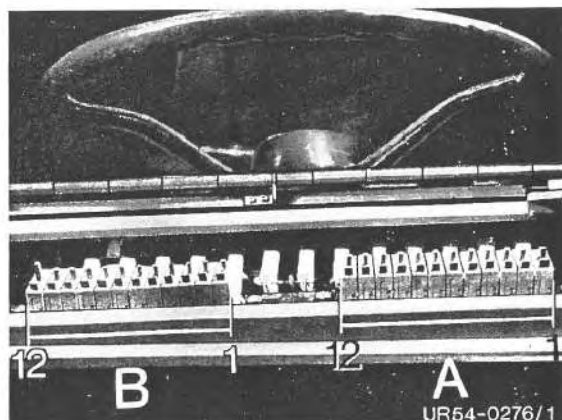
Electrolyte density corresponding to Baumé*	Specific weight*	Charged state of battery
32° Bé	1,285	well charged
24° Bé	1,20	half charged
16° Bé	1,12	discharged, charge immediately

* at an electrolyte temperature of +20°C

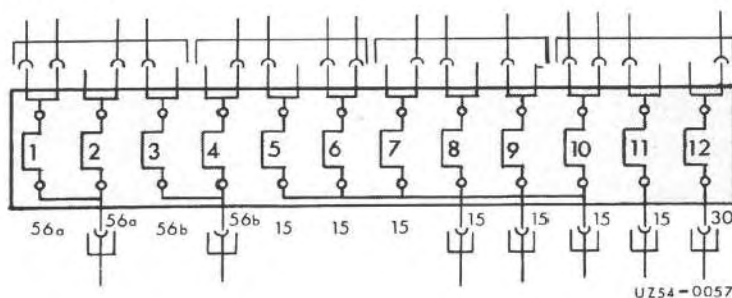
Electrolyte density (Tropics)

In 20°C		In 30°C		In 40°C		Charged status
Electrolyte density	Specific weight	Electrolyte density	Specific weight	Electrolyte density	Specific weight	
27,9° Bé	1,24	26,9° Bé	1,23	26° Bé	1,22	well charged
23°– 21° Bé	1,19 1,17	22°– 20° Bé	1,18 1,16	21°– 19° Bé	1,17– 1,15	half charged
14,2°– 10,6° Bé	1,11– 1,08	13°– 9,40° Bé	1,10– 1,07	11,2°– 8,0° Bé	1,09°– 1,06	discharged charge immediately

Automatic Cutouts (to chassis end No. 078 674)

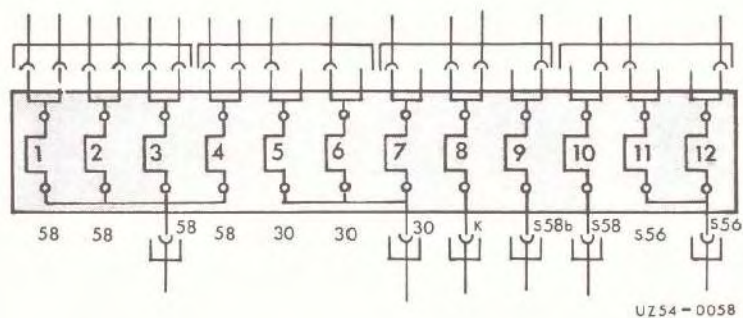


Automatic Cutouts A



No.	Terminal	A	Designation of power load
1	56 a	8	High beam, left, high beam indicator lamp
2	56 a	8	High beam, right
3	56 b	8	Low beam, left
4	56 b	8	Low beam, right
5	15	8	provided for connection of 3rd brake
6	15	8	Horn, call system with indicator lamp
7	15	8	Windscreen washer
8	15	8	Windscreen wiper combination switch
9	15	12	Heater fan
10	15	8	Indicator and warning lamp Supply pressure brake system, wear indicator, front and rear axle, four-wheel drive differential lock, parking brake, pressure difference indicator, charging indicator, reversing lamp, PTO shaft, instrument cluster (+), revolution counter terminal 1
11	15	12	Turn signal with indicator light, stop light, shift gate indicator, masked main light switch
12	30	8	Hazard warning flasher switch

Automatic Cutouts B

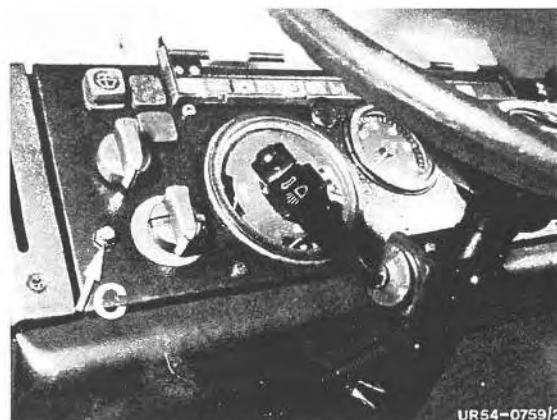
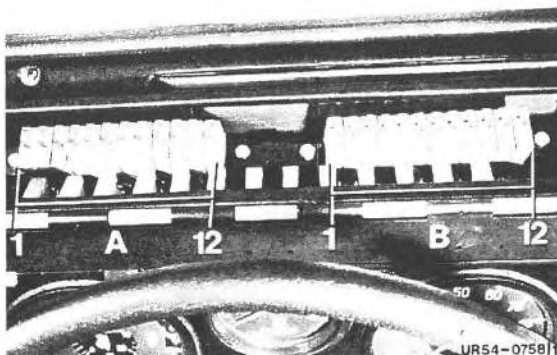


No.	Terminal	A	Designation of power load
1	58	8	Parking light, left
2	58	8	Relay shift gate indicator bright - dark, parking light, right
3	58	8	Tail light, left, rear fog lamp
4	58	8	Instrument illumination, tail light, right
5	30	8	Speedometer (1), reading light
6	30	8	Socket, instrument panel
7	30	8	12- pole trailer socket terminal 30
8	K	8	IR auxiliary light with indicator light
9	S 58 b	8	Convoy cross light
10	S 58	8	Masked tail light left and right, trailer socket 12-pole
11	S 56	8	Masked headlight, left
12	S 56	8	Masked headlight, right

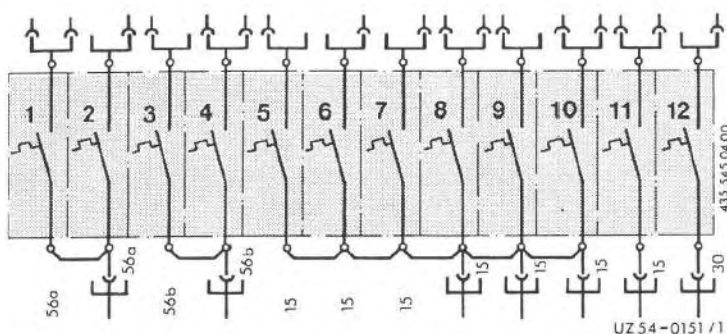
Automatic cutout C

No.	Terminal	A	Designation of power load
1	30	60	Power supply body

Automatic Cutouts (from chassis end No. 078 675)



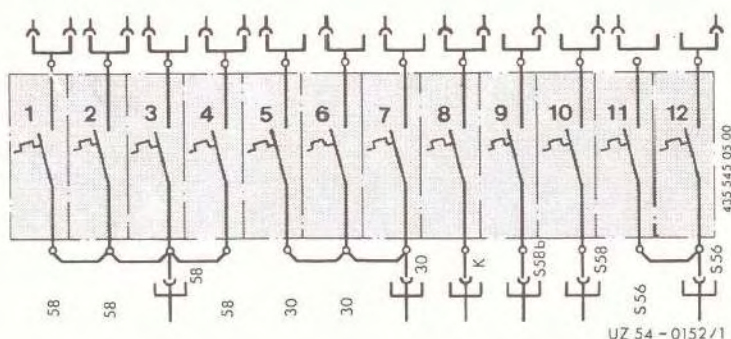
Automatic Cutouts A



Fuse Index ¹⁾	No.	Designation of power load	A
F 4	1	High beam, left, high beam indicator light	8
F 6	2	High beam, right	8
F 3	3	Low beam, left	8
F 5	4	Low beam, right	8
F10	5	Reserve	8
F 2	6	Call system (buzzer with indicator lamp)	8
F11	7	Windscreen washer	8
F12	8	Windscreen wiper (winch)	8
F13	9	Heater fan	12
F13	10	Indicator and warning lamps – Differential pressure gauge, brake wear indicator front - rear axle – Parking brake, four-wheel drive, differential lock – Supply pressure brake system, revolution counter, instrument cluster, PTO shaft	8
F 9	11	Turn signal, interior lamp, headlight flasher, shift gate indicator	8
F 8	12	Brake light, masked stop lamp, hazard warning system, horn, socket 12-pole	12

1) Designation in circuit diagram

Automatic Cutouts B



Fuse Index ¹⁾	No.	Designation of power load	A
F15	1	Parking light, left	8
F16	2	Parking light right, relay shift gate indicator (bright - dark)	8
F17	3	Tail light left, license plate lights, socket 12-pole	8
F18	4	Tail light right, instrument illumination	8
F19	5	Speedometer, map light	8
F20	6	Socket instrument panel	8
F21	7	Socket 2-pole, socket 12-pole	8
F 7	8	IR auxiliary light with indicator lamp	8
F22	9	Convoy cross light, socket 12-pole	8
F25	10	Masked tail light left and right, socket 12-pole	8
F23	11	Masked headlamp, left	8
F24	12	Masked headlamp, right	8

Automatic Cutout C

Fuse Index ¹⁾	No.	Designation of power load	A
F 1	—	Power supply body	60

1) Designation in circuit diagram

Bulbs in acc. with DIN 72 601

Power consumer	Output at 24 V W	Shape in acc. with DIN 72 601	Remarks
Main headlamps	55/50	A	
Supplementary headlamps for front attachments	55/50	A	
Masked headlamps	18	R	
Side light	4	HL	
Direction indicator front	21	RL	
Additional direction indicators front	4	HL	
Direction indicator rear	21	P 25-1	
Tail light	10	G	
Masked tail light	5	G	
Brake light	21	P 25-1	
Masked brake light	2	H	
Licence plate light (with masked brake circuit)	5	G	
Interior cab light	10	K	
Rotating beacon ¹⁾	45	U	
Rotating beacon ²⁾ (halogen)	70	H1	
Reversing light	35	D	
Convoy cross light	2	H	
Shift indicator light	1.2	W5/1.2	
Warning lights	2	H	
Instrument cluster, hazard warning switch, speedometer, symbol display, rotating beacon, heated windscreen			
Warning lights, additional headlamps for front attachments	4	HL	

1) Up to chassis end No. end April 1982

2) From chassis end No. May 1982

Electrical circuit diagram (up to chassis end No. 078 674)

Key to 1.5/2

1a	Direction indicator right	48	Diode box, masked circuit for hazard warning system
1b	Direction indicator left	49	Relay, shift indicator light/dark
2a	High beam/low beam right	50	Relay, hazard warning system
2b	High beam/low beam left	51	Hazard warning switch
3a	Side light right	52a	Isolating point, cab
3b	Side light left	52b	Isolating point, cab (not connected)
4a	Masked headlight right	53	Pickup, hazard warning system
4b	Masked headlight left	54	Generator
5	Infrared auxiliary lamp	55	Plug socket, power supply and calling system
6	Signal horn	56	Pickup, coolant temperature
7	Interior light, cab	57	Pickup, oil pressure gauge
8	Pickup for brake lining wear indicator, front axle left	58	Pickup, fuel gauge
9	Circuit-breaker, power supply system	59	Pickup, differential pressure gauge
10	Switch, rear fog light	60	Pickup, pto shaft
11	Switch, screen washer	61	Pickup, backup lamp
12	Dimmer, instrument light	62	Pickup, differential lock
13	Plug socket, 2-pin, instrument panel	63	Pickup, 4-wheel drive
14	Ignition switch	64	Pickup, parking brake
15	Speedometer	65	Pickup, shift indicator
16	Rev counter	66	Starter
17	Instrument cluster	67	Plug socket, external power supply
18	Switch, infrared auxiliary lamp	68	Main battery switch
19	Main/masked lights switch	69	Batteries
20	Reading lamp	70	Switch, calling system
21	Tell-tale, pto shaft	71	Pickup, brake lining wear indicator rear axle left
22	Tell-tale, infrared auxiliary lamp	72	Junction box in side member at rear left
23	Tell-tale, trailer direction indicators	73a	Direction indicator right
24	Tell-tale, tractor unit direction indicators	73b	Direction indicator left
25	Tell-tale, high beam	74a	Tail light right
26	Tell-tale, charging indicator	74b	Tail light left
27	Tell-tale, differential pressure gauge	75a	Stoplight right
28	Tell-tale, parking brake	75b	Stoplight left
29	Tell-tale, 4-wheel drive	76a	Masked tail light right
30	Tell-tale, differential lock VA/RA	76b	Masked tail light left
31	Tell-tale, brake lining wear indicator RA	77	Convey cross light
32	Tell-tale, brake lining wear indicator FA	78	Plug socket 7-pin ¹⁾
33	Tell-tale, tank pressure	79	Plug socket 12-pin
34	Tell-tale, calling system	80	Plug socket 2-pin
35	Tell-tale, shift indicator	81	Backup lamp ¹⁾
36	Isolating relay, power supply	82	Rear fog lamp ¹⁾
37	Power supply relay	83	Masked stop light
38	Diode group	84a	Licence plate light right
39	Wiper motor	84b	Licence plate light left
40	Blower		
41	Pump, screen washer		
42	Control switch for blower		
43	Combination switch		
44	Regulator		
45	Stoplight pickup		
46a	Automatic cutout A		
46b	Automatic cutout B		
47	Buzzer, calling system		

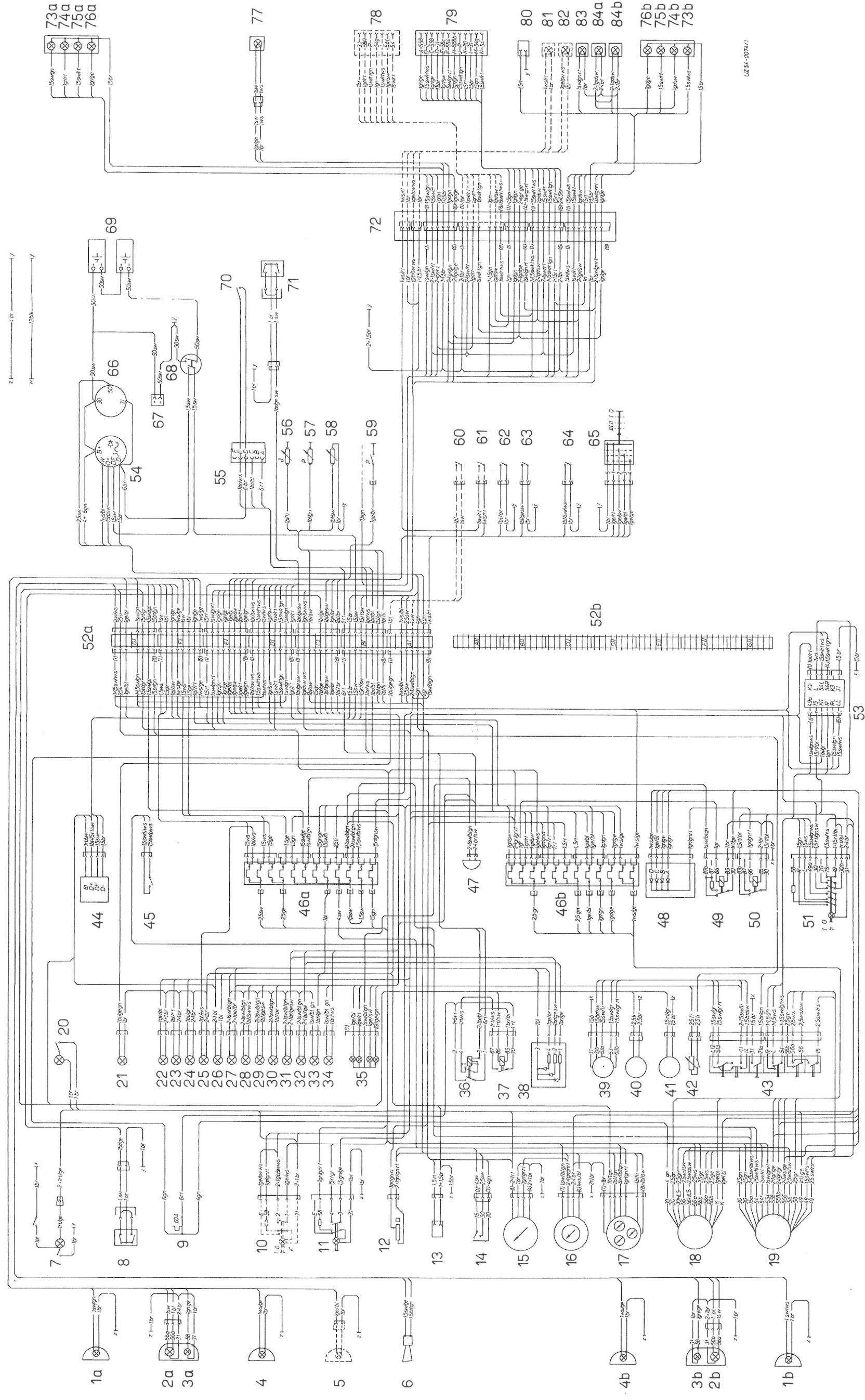
- 1) Retrofitting possible
.... Special version

m = Ground connection engine
X = Ground connection instrument panel
Y = Ground connection frame
Z = Ground connection cab

Line colour code:

bl = blue
blk = blank
br = brown
ge = yellow
gn = green
gr = grey
li = purple
rs = pink
rt = red
sw = black
ws = white

Electrical circuit diagram (up to chassis end No. 078 674)



Designations in switching and circuit diagrams**Target designation:**

The target designation refers to the current path on which the line continues.

Example: → 15= Continuation of line on current path 15

Designation of the plug connectors

Al- Gl= plug-and-socket connection driver's cab

Example: Plug - and - socket connection Al/1

Identification of plug-and-socket connection= Al

Identification of connection=/1

Identification of ground connection points

m = Ground engine
X = Ground instrument panel
y = Ground frame
z = Ground driver's cab

Line colour code

bl = blue
blk = bright
br = brown
ge = yellow
gn = green
gr = grey
li = violet
nf = natural coloured
rs = pink
rt = red
sw = black
ws = white

Example: Line designation 1.5 gr/rt

Line cross - section- 1.5 mm²

Basic colour gr- grey

Colour code rt- red

Circuit Diagram (from chassis end No. 078 765)

Keys to 1.5/6

1 Power Supply, Starting Control

- G 1, 2 Batteries
- G 3 Alternator
- H 1 Indicator lamp charging indicator
- M 1 Starter
- P 1 Governor
- Q 1 Main battery switch
- Q 2 Drive switch
- X 1 Assisted start socket

2 Power Supply and Call System Structure

- F 1 Overcurrent protection switch, power supply
- F 2 Overcurrent protection switch, call system
- H 2 Indicator lamp, call system
- H 3 Buzzer, call system
- K 1 Isolating relay, power supply
- K 2 Relay, power supply
- S 1 Rocker switch, call system
- X 2 Socket, power supply and call system

3 Headlamp, IR light, brake system

- F 3 Overcurrent protection switch, low beam left
- F 4 Overcurrent protection switch, high beam left
- High beam indicator
- F 5 Overcurrent protection switch, low beam right, rear fog light
- F 6 Overcurrent protection switch, high beam right
- F 7 Overcurrent protection switch, IR auxiliary light and indicator lamp
- F 8 Overcurrent protection switch, stop lamps
- Masked stop lamps, horn, turn signal and hazard warning system
- H 4 Headlamp, left
- H 5 Indicator lamp, high beam
- H 6 Headlamp, right
- H 7 Indicator lamp IR auxiliary light
- H 8 IR auxiliary light
- H 9 Stop light, right
- H 10 Stop light, left
- H 11 Masked stop lamp
- S 2 High/low beam switch
- S 3 Switch IR auxiliary light
- S 4 Sending unit stop light
- S 5 Main/masked light switch
- X 3 Socket 7-pole*
- X 4 Socket 12-pole*

4 Headlight Flasher, Interior Lamp, Reversing Lamp*

- F 9 Overcurrent protection switch, shift gate indicator
- Reversing lamp, headlight flasher interior lamp
- H 12 Reversing lamp*
- H 13 Interior lamp
- K 3 Relay masked light
- S 6 Headlight flasher
- S 7 Sending unit reversing lamp
- S 8, 9 Switch interior lamp

5 Turn Signal Hazard Warning System

- H 14 Indicator lamp, turn signal, tractor
- H 15 Turn signal, rear left
- H 16 Turn signal, front left
- H 17 Turn signal, front right
- H 18 Turn signal, rear right
- H 19 Indicator lamp, turn signal trailer
- H 20 Indicator lamp, hazard warning system
- P 2 Sending unit, hazard warning system
- S 10 Hazard warning flasher switch
- S 11 Turn signal switch
- X 3 Socket 7-pole*
- X 4 Socket 12-pole

6 Horn

- H 21 Horn
- S 12 Switch, horn
- X 5 Connection, alarm switch

7 Wiper and Washer Unit

- F 10 Overcurrent protection switch, reserve
- F 11 Overcurrent protection switch, windscreen washer
- F 12 Overcurrent protection switch, windscreen wiper
- M 2 Windscreen washer pump
- M 3 Wiper motor
- S 13 Switch, windscreen washer
- S 14 Switch, windscreen wiper
- A= A= ON-OFF, B= slow -fast

8 Heater

- F 13 Overcurrent protection switch heater fan
- M 4 Heater fan
- S 15 Switch, heater fan

9 Operation Monitoring

- A 1 Diode group
- B 1 Sending unit, coolant temperature indicator
- B 2 Sending unit, oil pressure gauge
- B 3 Sending unit, fuel gauge
- F 14 Overcurrent protection switch, instruments indicator lamps
- H 22 Indicator lamp, differential pressure gauge
- H 23 Indicator lamp, brake pad wear indicator RA
- H 24 Indicator lamp, brake pad wear indicator FA
- H 25 Indicator lamp, parking brake
- H 26 Indicator lamp, four-wheel drive
- H 27 Indicator lamp, differential lock
- FA/RA
- H 28 Indicator lamp supply pressure
- H 29 Revolution counter
- H 30 Instrument cluster
- S 16 Sending unit, differential pressure gauge
- S 17 Sending unit, brake pad wear indicator RA
- S 18 Sending unit, brake pad wear indicator FA

Key to 1.5/6 (Continued)

S 19 Sending unit, parking brake
 S 20 Sending unit, four - wheel drive
 S 21 Sending unit, differential lock
 V 1 Diode, differential pressure gauge
 V 2 Diode, brake pad wear indicator RA
 V 3 Diode, brake pad wear indicator FA

H 44 Convoy cross light
 H 45 Masked light - headlamp, left
 H 46 Masked light - headlamp, right
 H 47 Masked light - tail light, right
 H 48 Masked light - tail light, left
 X 4 Socket 12-pole

**10 Side-marker Lamp, License Plate Light
Tail Light, Instrument Illumination**

F 15 Overcurrent protection switch, side-marker lamp left, indicator lamp - rear fog light
 F 16 Overcurrent protection switch, side-marker lamp right, relay shift gate indicator
 F 17 Overcurrent protection switch license plate light, tail light left
 F 18 Overcurrent protection switch instrument illumination, tail light right
 H 31 Side-marker lamp, left
 H 32 Rear fog light*
 H 33 Side-marker lamp, right
 H 34-37 Shift gate indicator
 H 38, 39 License plate, light
 H 40 Tail light, left
 H 41 Tail light, right
 K 4 Relay shift gate indicator (bright -dark)
 S 22 Switch, rear fog light*
 S 23 Sending unit, shift gate indicator
 S 24 Regulating switch instrument illumination
 X 3 Socket 7-pole*
 X 4 Socket 12-pole

13 Sockets

X 3 Socket, 7-pole
 X 4 Socket, 12-pole

14 Recovery winch

K 5 Electropneumatic pilot control valve
 S 26 Button, recovery winch
 X 8 Socket, recovery winch

AI-GI = Plug-and-socket connection, driver's cab
 VD = Plug-and-socket connection, distributor box rear
 X = Ground bulkhead
 Y = Ground frame
 Z = Ground driver's cab

* Installation possible

11 Speedometer, Map Light, Sockets

F 19 Overcurrent protection switch, speedometer map light
 F 20 Overcurrent protection switch socket driver's cab
 F 21 Overcurrent protection switch socket 2-pole
 H 42 Speedometer
 H 43 Map light
 S 25 Switch, map light
 X 4 Socket, 12-pole
 X 6 Socket, driver's cab
 X 7 Socket, 2-pole

12 Masked light

F 22 Overcurrent protection switch, convoy cross light
 F 23 Overcurrent protection switch, masked headlight left
 F 24 Overcurrent protection switch, masked headlight right
 F 25 Overcurrent protection switch, masked tail light

Circuit Diagram (from chassis end No. 078 675)

